

Revision Questions and Detailed Solutions

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1 Question 1: Transmission Lines

Original Question

The input impedance Z_{in} , terminated by a load impedance Z_L , for a lossless transmission line of length l is

$$Z_{\text{in}} = Z_0 \frac{Z_L + jZ_0 \tan(\beta l)}{Z_0 + jZ_L \tan(\beta l)}. \quad (1)$$

- a) Determine the electrical length of a 1.5 m lossless transmission line if the phase constant is 4 rad/m.
- b) Interpret the expression for Z_{in} when the load impedance Z_L is:
 - i. a matched load;
 - ii. a short circuit;
 - iii. an open circuit.
- c) If the characteristic impedance and load impedance are

$$Z_0 = 50 \Omega, \quad Z_L = 83 - j25 \Omega,$$

calculate:

- i. the absolute load impedance;
 - ii. the load reflection coefficient;
 - iii. the load VSWR;
 - iv. the load return loss.
- d) An inductance of 88 nH is to be created in a 50 Ω lossless microstrip line by using a short-circuited stub at a frequency of 1.2 GHz. If the propagation velocity is 2.7×10^8 m/s, calculate the required stub length.

Detailed Solution

For a lossless transmission line, the input impedance is

$$Z_{\text{in}} = Z_0 \frac{Z_L + jZ_0 \tan(\beta l)}{Z_0 + jZ_L \tan(\beta l)}. \quad (2)$$

Here, Z_0 is the characteristic impedance, Z_L is the load impedance, β is the phase constant, and l is the physical length of the transmission line.

1.1 Electrical length

Given

$$l = 1.5 \text{ m}, \quad \beta = 4 \text{ rad/m}, \quad (3)$$

the electrical length is

$$\theta = \beta l. \quad (4)$$

Therefore,

$$\theta = (4 \text{ rad/m}) (1.5 \text{ m}) \quad (5)$$

$$= 6 \text{ rad}. \quad (6)$$

In degrees,

$$\theta = 6 \left(\frac{180^\circ}{\pi} \right) \quad (7)$$

$$= 343.77^\circ. \quad (8)$$

Hence,

$$\boxed{\theta = 6 \text{ rad} = 343.77^\circ.} \quad (9)$$

1.2 Special load conditions

1.2.1 Matched load

For a matched load,

$$Z_L = Z_0. \quad (10)$$

Substituting into the transmission-line equation,

$$Z_{\text{in}} = Z_0 \frac{Z_0 + jZ_0 \tan(\beta l)}{Z_0 + jZ_0 \tan(\beta l)} \quad (11)$$

$$= Z_0. \quad (12)$$

Thus,

$$\boxed{Z_{\text{in}} = Z_0.} \quad (13)$$

A matched line has no reflected wave, so the input impedance is independent of line length.

1.2.2 Short-circuit load

For a short circuit,

$$Z_L = 0. \quad (14)$$

Then

$$Z_{\text{in}} = Z_0 \frac{jZ_0 \tan(\beta l)}{Z_0} \quad (15)$$

$$= jZ_0 \tan(\beta l). \quad (16)$$

Therefore,

$$\boxed{Z_{\text{in}} = jZ_0 \tan(\beta l).} \quad (17)$$

The input is purely reactive. For a short stub shorter than $\lambda/4$, the input behaves inductively.

1.2.3 Open-circuit load

For an open circuit,

$$Z_L \rightarrow \infty. \quad (18)$$

Dividing the numerator and denominator by Z_L ,

$$Z_{\text{in}} = Z_0 \frac{1 + j(Z_0/Z_L) \tan(\beta l)}{(Z_0/Z_L) + j \tan(\beta l)} \quad (19)$$

$$= \frac{Z_0}{j \tan(\beta l)} \quad (20)$$

$$= -jZ_0 \cot(\beta l). \quad (21)$$

Hence,

$$\boxed{Z_{\text{in}} = -jZ_0 \cot(\beta l).} \quad (22)$$

The open-circuit stub is also purely reactive.

1.3 Load calculations for $Z_0 = 50 \Omega$ and $Z_L = 83 - j25 \Omega$

1.3.1 Absolute load impedance

$$|Z_L| = \sqrt{83^2 + (-25)^2} \quad (23)$$

$$= \sqrt{7514} \quad (24)$$

$$= 86.68 \Omega. \quad (25)$$

Therefore,

$$\boxed{|Z_L| = 86.68 \Omega.} \quad (26)$$

The phase of the impedance is

$$\angle Z_L = \tan^{-1} \left(\frac{-25}{83} \right) \quad (27)$$

$$= -16.77^\circ, \quad (28)$$

so

$$Z_L = 86.68 \angle -16.77^\circ \Omega. \quad (29)$$

1.3.2 Load reflection coefficient

The load reflection coefficient is

$$\Gamma_L = \frac{Z_L - Z_0}{Z_L + Z_0}. \quad (30)$$

Substituting,

$$\Gamma_L = \frac{(83 - j25) - 50}{(83 - j25) + 50} \quad (31)$$

$$= \frac{33 - j25}{133 - j25}. \quad (32)$$

Multiplying numerator and denominator by the complex conjugate $133 + j25$,

$$\Gamma_L = \frac{(33 - j25)(133 + j25)}{133^2 + 25^2} \quad (33)$$

$$= 0.2738 - j0.1365. \quad (34)$$

Its magnitude is

$$|\Gamma_L| = \sqrt{0.2738^2 + (-0.1365)^2} \quad (35)$$

$$= 0.3059, \quad (36)$$

and its phase is

$$\angle \Gamma_L = \tan^{-1} \left(\frac{-0.1365}{0.2738} \right) \quad (37)$$

$$= -26.50^\circ. \quad (38)$$

Therefore,

$$\boxed{\Gamma_L = 0.2738 - j0.1365 = 0.3059 \angle -26.50^\circ.} \quad (39)$$

1.3.3 Voltage standing-wave ratio

$$\text{VSWR} = \frac{1 + |\Gamma_L|}{1 - |\Gamma_L|}. \quad (40)$$

Thus,

$$\text{VSWR} = \frac{1 + 0.3059}{1 - 0.3059} \quad (41)$$

$$= 1.8815. \quad (42)$$

Hence,

$$\boxed{\text{VSWR} \approx 1.88 : 1.} \quad (43)$$

1.3.4 Return loss

$$RL = -20 \log_{10} |\Gamma_L|. \quad (44)$$

Therefore,

$$RL = -20 \log_{10}(0.3059) \quad (45)$$

$$= 10.29 \text{ dB}. \quad (46)$$

Hence,

$$\boxed{RL = 10.29 \text{ dB}.} \quad (47)$$

1.4 Short-circuited stub used as an inductor

For a short-circuited lossless stub,

$$Z_{\text{in}} = jZ_0 \tan(\beta l). \quad (48)$$

To emulate an inductance L ,

$$Z_{\text{in}} = j\omega L. \quad (49)$$

Equating the two expressions,

$$Z_0 \tan(\beta l) = \omega L. \quad (50)$$

Hence,

$$\tan(\beta l) = \frac{\omega L}{Z_0}. \quad (51)$$

Given

$$L = 88 \text{ nH}, \quad Z_0 = 50 \Omega, \quad f = 1.2 \text{ GHz}, \quad v_p = 2.7 \cdot 10^8 \text{ m/s}, \quad (52)$$

$$\omega = 2\pi f = 2\pi(1.2 \times 10^9) \quad (53)$$

$$= 7.5398 \times 10^9 \text{ rad/s}. \quad (54)$$

Then

$$\frac{\omega L}{Z_0} = \frac{(7.5398 \times 10^9)(88 \times 10^{-9})}{50} \quad (55)$$

$$= 13.2701. \quad (56)$$

Therefore,

$$\beta l = \tan^{-1}(13.2701) \quad (57)$$

$$= 1.4956 \text{ rad} \quad (58)$$

$$= 85.69^\circ. \quad (59)$$

The phase constant is

$$\beta = \frac{\omega}{v_p} = \frac{2\pi f}{v_p} \quad (60)$$

$$= \frac{2\pi(1.2 \times 10^9)}{2.7 \times 10^8} \quad (61)$$

$$= 27.9253 \text{ rad/m.} \quad (62)$$

Hence,

$$l = \frac{1.4956}{27.9253} \quad (63)$$

$$= 0.05356 \text{ m.} \quad (64)$$

Therefore,

$$\boxed{l = 53.56 \text{ mm.}} \quad (65)$$

The shortest positive solution is used. Other solutions differ by integer multiples of $\lambda/2$.

2 Question 2: Ideal Directional Coupler

Original Question

Figures 1(a) and 1(b) in the question paper illustrate an ideal 10 dB directional coupler and its equivalent scattering matrix. The phase is $\phi = 90^\circ$, and, for excitation at port 1,

$$V_1 = V, \quad V_2 = -j\sqrt{1 - k^2} V, \quad V_3 = 0, \quad V_4 = kV.$$

Interpret the values of the scattering-matrix elements:

- (i) if all four ports are perfectly matched;
- (ii) if all four ports are symmetrical;
- (iii) if ports 1 and 3, and ports 2 and 4, are isolated from one another;
- (iv) and hence write the full S -matrix.

Detailed Solution

A 10 dB directional coupler has voltage coupling coefficient

$$k = 10^{-C_{dB}/20} \tag{66}$$

$$= 10^{-10/20} \tag{67}$$

$$= 0.3162. \tag{68}$$

The through coefficient is

$$t = \sqrt{1 - k^2} \tag{69}$$

$$= \sqrt{1 - 0.3162^2} \tag{70}$$

$$= 0.9487. \tag{71}$$

For excitation at port 1,

$$V_1 = V, \quad V_2 = -jtV, \quad V_3 = 0, \quad V_4 = kV. \tag{72}$$

2.1 All four ports perfectly matched

Perfect matching means that no wave is reflected back into the same port. Therefore,

$$\boxed{S_{11} = S_{22} = S_{33} = S_{44} = 0.} \tag{73}$$

2.2 All four ports symmetrical

For a reciprocal, symmetrical four-port coupler, corresponding transmission coefficients are equal:

$$S_{12} = S_{21} = S_{34} = S_{43}, \quad (74)$$

and

$$S_{14} = S_{41} = S_{23} = S_{32}. \quad (75)$$

For the stated phase convention,

$$S_{12} = S_{21} = S_{34} = S_{43} = -j\sqrt{1-k^2}, \quad (76)$$

and

$$S_{14} = S_{41} = S_{23} = S_{32} = k. \quad (77)$$

2.3 Isolated port pairs

The isolated port pairs are ports 1 and 3, and ports 2 and 4. Thus,

$$\boxed{S_{13} = S_{31} = S_{24} = S_{42} = 0.} \quad (78)$$

2.4 Full scattering matrix

The full ideal scattering matrix is

$$[S] = \begin{bmatrix} 0 & -j\sqrt{1-k^2} & 0 & k \\ -j\sqrt{1-k^2} & 0 & k & 0 \\ 0 & k & 0 & -j\sqrt{1-k^2} \\ k & 0 & -j\sqrt{1-k^2} & 0 \end{bmatrix}. \quad (79)$$

Using $k = 0.3162$ and $\sqrt{1-k^2} = 0.9487$,

$$\boxed{[S] = \begin{bmatrix} 0 & -j0.9487 & 0 & 0.3162 \\ -j0.9487 & 0 & 0.3162 & 0 \\ 0 & 0.3162 & 0 & -j0.9487 \\ 0.3162 & 0 & -j0.9487 & 0 \end{bmatrix}.} \quad (80)$$

This matrix is matched, reciprocal, lossless, symmetrical, and directional.

2.5 Directional-coupler port layout

Figure 1 shows the port convention consistent with the given excitation. Port 1 is the input port, port 2 is the through port, port 3 is the isolated port, and port 4 is the coupled port.

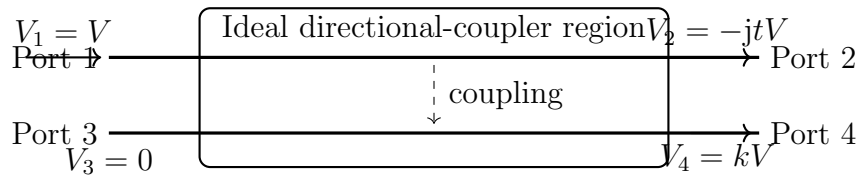


Figure 1: Port layout of the ideal directional coupler used in Question 2.

3 Question 3: Directional Coupler and Wilkinson Divider

Original Question

Part A: Coupled-line directional coupler

A matched 12 dB microstrip directional coupler is fabricated on a substrate with relative permittivity $\varepsilon_r = 12$ and thickness $h = 0.19$ mm. All ports are terminated in $50\ \Omega$. Determine:

- (i) the coupling coefficient k ;
- (ii) the even-mode and odd-mode characteristic impedances;
- (iii) the widths of the even-mode and odd-mode microstrip lines.

Part B: Wilkinson power divider

A 3 dB Wilkinson power splitter is designed to operate at 2.4 GHz. The substrate has $\varepsilon_r = 9.8$ and $h = 1.88$ mm, and all port impedances are $50\ \Omega$. Determine:

- (i) the impedances of the quarter-wave sections;
- (ii) the widths of the input and output lines;
- (iii) the width of the quarter-wave transformer;
- (iv) the effective dielectric constant of the transformer;
- (v) the physical length of the quarter-wave transformer.

Detailed Solution

3.1 Part A: 12 dB coupled-line directional coupler

Given

$$C = 12\ \text{dB}, \quad Z_0 = 50\ \Omega, \quad \varepsilon_r = 12, \quad h = 0.19\ \text{mm}. \quad (81)$$

3.1.1 Coupling coefficient

$$k = 10^{-C/20} \quad (82)$$

$$= 10^{-12/20} \quad (83)$$

$$= 0.2512. \quad (84)$$

Therefore,

$$\boxed{k = 0.2512.} \quad (85)$$

3.1.2 Even-mode and odd-mode impedances

For a symmetrical coupled-line coupler,

$$Z_{0e} = Z_0 \sqrt{\frac{1+k}{1-k}}, \quad (86)$$

and

$$Z_{0o} = Z_0 \sqrt{\frac{1-k}{1+k}}. \quad (87)$$

Thus,

$$Z_{0e} = 50 \sqrt{\frac{1+0.2512}{1-0.2512}} \quad (88)$$

$$= 64.63 \, \Omega, \quad (89)$$

and

$$Z_{0o} = 50 \sqrt{\frac{1-0.2512}{1+0.2512}} \quad (90)$$

$$= 38.68 \, \Omega. \quad (91)$$

Therefore,

$$\boxed{Z_{0e} = 64.63 \, \Omega, \quad Z_{0o} = 38.68 \, \Omega.} \quad (92)$$

A useful check is

$$\sqrt{Z_{0e} Z_{0o}} = \sqrt{(64.63)(38.68)} \quad (93)$$

$$\approx 50 \, \Omega. \quad (94)$$

3.1.3 Approximate microstrip widths

The Hammerstad inversion is used to estimate the microstrip width. First,

$$A = \frac{Z_0}{60} \sqrt{\frac{\varepsilon_r + 1}{2}} + \frac{\varepsilon_r - 1}{\varepsilon_r + 1} \left(0.23 + \frac{0.11}{\varepsilon_r} \right). \quad (95)$$

For $w/h \leq 2$,

$$\frac{w}{h} = \frac{8e^A}{e^{2A} - 2}. \quad (96)$$

For the even-mode impedance,

$$\left(\frac{w}{h} \right)_e \approx 0.4216. \quad (97)$$

Therefore,

$$w_e = 0.4216(0.19) \quad (98)$$

$$= 0.0801 \text{ mm}. \quad (99)$$

For the odd-mode impedance,

$$\left(\frac{w}{h}\right)_o \approx 1.3292. \quad (100)$$

Therefore,

$$w_o = 1.3292(0.19) \quad (101)$$

$$= 0.2526 \text{ mm}. \quad (102)$$

Hence,

$$\boxed{w_e \approx 0.080 \text{ mm}, \quad w_o \approx 0.253 \text{ mm}.} \quad (103)$$

These are first-order uncoupled microstrip estimates. A practical edge-coupled coupler must also determine the line spacing, and an EM simulator is normally used for final optimisation because even-mode and odd-mode impedances depend on both w/h and s/h .

3.2 Part B: 3 dB Wilkinson power divider

Given

$$f_0 = 2.4 \text{ GHz}, \quad Z_0 = 50 \Omega, \quad \varepsilon_r = 9.8, \quad h = 1.88 \text{ mm}. \quad (104)$$

3.2.1 Quarter-wave branch impedances

For an equal-split Wilkinson divider,

$$Z_{\lambda/4} = \sqrt{2}Z_0. \quad (105)$$

Thus,

$$Z_{\lambda/4} = \sqrt{2}(50) \quad (106)$$

$$= 70.71 \Omega. \quad (107)$$

The isolation resistor is

$$R = 2Z_0 = 100 \Omega. \quad (108)$$

Therefore,

$$\boxed{Z_{\lambda/4} = 70.71 \Omega, \quad R = 100 \Omega.} \quad (109)$$

3.2.2 Widths of the input and output lines

The input and output lines are $50\ \Omega$ lines. Using the Hammerstad inversion,

$$\frac{w_{50}}{h} \approx 0.9752. \quad (110)$$

Therefore,

$$w_{50} = 0.9752(1.88) \quad (111)$$

$$= 1.833\text{ mm}. \quad (112)$$

Hence,

$$\boxed{w_{\text{in}} = w_{\text{out}} \approx 1.83\text{ mm}.} \quad (113)$$

3.2.3 Width of each quarter-wave transformer

For $Z_{\lambda/4} = 70.71\ \Omega$,

$$\frac{w_t}{h} \approx 0.4274. \quad (114)$$

Thus,

$$w_t = 0.4274(1.88) \quad (115)$$

$$= 0.803\text{ mm}. \quad (116)$$

Therefore,

$$\boxed{w_t \approx 0.803\text{ mm}.} \quad (117)$$

3.2.4 Effective dielectric constant

For $w/h < 1$, a common approximation is

$$\varepsilon_{\text{eff}} = \frac{\varepsilon_r + 1}{2} + \frac{\varepsilon_r - 1}{2} \left(1 + \frac{12h}{w} \right)^{-1/2}. \quad (118)$$

Using $w/h = 0.4274$,

$$\varepsilon_{\text{eff}} = \frac{9.8 + 1}{2} + \frac{9.8 - 1}{2} \left(1 + \frac{12}{0.4274} \right)^{-1/2} \quad (119)$$

$$= 6.216. \quad (120)$$

Therefore,

$$\boxed{\varepsilon_{\text{eff}} \approx 6.216.} \quad (121)$$

3.2.5 Physical length of the quarter-wave transformer

The guided wavelength is

$$\lambda_g = \frac{c}{f_0 \sqrt{\varepsilon_{\text{eff}}}}. \quad (122)$$

Thus,

$$\lambda_g = \frac{3.0 \times 10^8}{(2.4 \times 10^9) \sqrt{6.216}} \quad (123)$$

$$= 50.14 \text{ mm}. \quad (124)$$

Therefore,

$$l_{\lambda/4} = \frac{\lambda_g}{4} \quad (125)$$

$$= 12.53 \text{ mm}. \quad (126)$$

Hence,

$$l_{\lambda/4} \approx 12.53 \text{ mm}. \quad (127)$$

3.2.6 Wilkinson power-divider layout

The calculated dimensions can be arranged as shown in Figure 2. Each branch is a 70.71Ω , quarter-wave microstrip section. The 100Ω resistor connects the two output ports and provides isolation when the output signals are unequal.

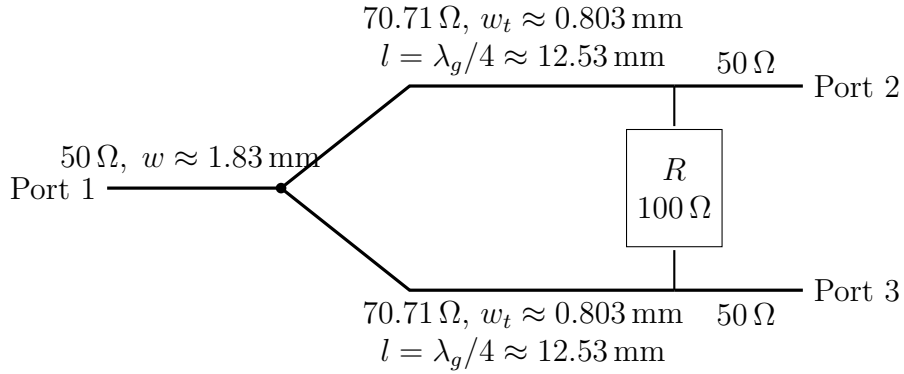


Figure 2: Layout of the equal-split Wilkinson power divider.

4 Question 4: Quarter-Wave Transformers

Original Question

Design a microstrip quarter-wave transformer to match a $50\ \Omega$ microwave source to a $100\ \Omega$ planar antenna at 3 GHz. The substrate has relative permittivity $\varepsilon_r = 9.8$ and thickness $h = 0.9\text{ mm}$. Calculate:

- (i) the transformer characteristic impedance Z_{0T} ;
- (ii) the transformer width w ;
- (iii) the effective dielectric constant ε_{eff} ;
- (iv) the guided wavelength λ_g ;
- (v) the physical transformer length;
- (vi) and sketch the circuit layout.

Using the same source, load, substrate, and centre frequency, design a maximally flat three-section quarter-wave transformer. The adjacent section impedances satisfy

$$\ln\left(\frac{Z_{n+1}}{Z_n}\right) = 2^{-N} \binom{N}{n} \ln\left(\frac{Z_L}{Z_0}\right), \quad (128)$$

where

$$N = \text{number of transformer sections}, \quad \binom{N}{n} = \frac{N!}{(N-n)!n!},$$

Z_n is the impedance of the n -th step, Z_L is the terminating impedance, and Z_0 is the source impedance.

Detailed Solution

4.1 Single-section transformer

The source impedance and load impedance are

$$Z_0 = 50\ \Omega, \quad Z_L = 100\ \Omega. \quad (129)$$

The operating frequency and substrate parameters are

$$f_0 = 3\text{ GHz}, \quad \varepsilon_r = 9.8, \quad h = 0.9\text{ mm}. \quad (130)$$

4.1.1 Transformer impedance

For a quarter-wave transformer,

$$Z_{0T} = \sqrt{Z_0 Z_L}. \quad (131)$$

Therefore,

$$Z_{0T} = \sqrt{(50)(100)} \quad (132)$$

$$= 70.71 \, \Omega. \quad (133)$$

Hence,

$$\boxed{Z_{0T} = 70.71 \, \Omega.} \quad (134)$$

4.1.2 Transformer width

For $\varepsilon_r = 9.8$ and $Z_{0T} = 70.71 \, \Omega$,

$$\frac{w}{h} \approx 0.4274. \quad (135)$$

Therefore,

$$w = 0.4274(0.9) \quad (136)$$

$$= 0.3846 \, \text{mm}. \quad (137)$$

Thus,

$$\boxed{w \approx 0.385 \, \text{mm}.} \quad (138)$$

4.1.3 Effective dielectric constant

$$\varepsilon_{\text{eff}} = \frac{9.8 + 1}{2} + \frac{9.8 - 1}{2} \left(1 + \frac{12}{0.4274} \right)^{-1/2} \quad (139)$$

$$= 6.216. \quad (140)$$

Hence,

$$\boxed{\varepsilon_{\text{eff}} \approx 6.216.} \quad (141)$$

4.1.4 Guided wavelength

$$\lambda_g = \frac{c}{f_0 \sqrt{\varepsilon_{\text{eff}}}} \quad (142)$$

$$= \frac{3.0 \times 10^8}{(3.0 \times 10^9) \sqrt{6.216}} \quad (143)$$

$$= 40.11 \, \text{mm}. \quad (144)$$

Therefore,

$$\boxed{\lambda_g \approx 40.11 \, \text{mm}.} \quad (145)$$

4.1.5 Transformer length

$$l = \frac{\lambda_g}{4} \quad (146)$$

$$= \frac{40.11}{4} \quad (147)$$

$$= 10.03 \text{ mm}. \quad (148)$$

Hence,

$$\boxed{l \approx 10.03 \text{ mm}.} \quad (149)$$

4.1.6 Circuit layout

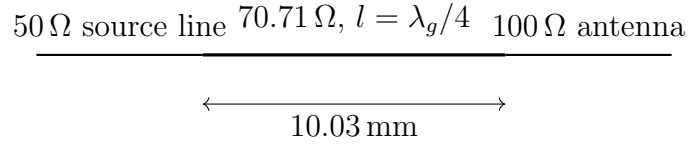


Figure 3: Single-section quarter-wave transformer.

4.2 Three-section maximally flat binomial transformer

For an $N = 3$ binomial transformer,

$$\ln \left(\frac{Z_{n+1}}{Z_n} \right) = 2^{-N} \binom{N}{n} \ln \left(\frac{Z_L}{Z_0} \right), \quad n = 0, 1, 2, 3. \quad (150)$$

Here,

$$\frac{Z_L}{Z_0} = \frac{100}{50} = 2, \quad \ln 2 = 0.69315. \quad (151)$$

The binomial coefficients are

$$\binom{3}{0} = 1, \quad \binom{3}{1} = 3, \quad \binom{3}{2} = 3, \quad \binom{3}{3} = 1. \quad (152)$$

The first section is

$$\ln \left(\frac{Z_1}{50} \right) = \frac{1}{8} \ln 2, \quad (153)$$

$$Z_1 = 50(2)^{1/8} \quad (154)$$

$$= 54.53 \Omega. \quad (155)$$

The second section is

$$\ln \left(\frac{Z_2}{Z_1} \right) = \frac{3}{8} \ln 2, \quad (156)$$

$$Z_2 = Z_1(2)^{3/8} \quad (157)$$

$$= 70.71 \Omega. \quad (158)$$

The third section is

$$\ln \left(\frac{Z_3}{Z_2} \right) = \frac{3}{8} \ln 2, \quad (159)$$

$$Z_3 = Z_2(2)^{3/8} \quad (160)$$

$$= 91.70 \, \Omega. \quad (161)$$

Finally,

$$Z_L = Z_3(2)^{1/8} \quad (162)$$

$$= 100 \, \Omega, \quad (163)$$

which confirms the design.

The corresponding approximate dimensions are summarised below.

Table 1: Three-section binomial transformer dimensions.

Section	$Z_n \, (\Omega)$	w/h	$w \, (\text{mm})$	$l = \lambda_g/4 \, (\text{mm})$
1	54.53	0.8115	0.730	9.80
2	70.71	0.4274	0.385	10.03
3	91.70	0.1887	0.170	10.25

Thus,

$$\boxed{Z_1 = 54.53 \, \Omega, \quad Z_2 = 70.71 \, \Omega, \quad Z_3 = 91.70 \, \Omega.} \quad (164)$$

The physical lengths are slightly different because each section has a different effective dielectric constant.

5 Question 5: Oscillators and Varactor Diode

Original Question

- a) Explain the principles of the Nyquist criterion for oscillation. Also explain how the Colpitts-oscillator feedback network can be used to select suitable reactive components, namely inductors and capacitors, that contribute to energy efficiency and environmental sustainability in modern wireless communication systems.
- b) Calculate the oscillation frequency of a Colpitts oscillator using

$$L = 70 \text{ nH}, \quad C_1 = 60 \text{ pF}, \quad C_2 = 350 \text{ pF}.$$

- c) Calculate C_1 for a Colpitts oscillator if $C_2 = 7 \text{ nF}$ and the transistor base-current gain is $\beta = 45$.
- d) Calculate the inductance L required for a Colpitts oscillator to operate at 30 MHz using

$$C_1 = 170 \text{ pF}, \quad C_2 = 190 \text{ pF}.$$

- e) A silicon graded-junction PN diode has a junction capacitance of 22 pF at zero bias. Determine the reverse-bias voltage required to obtain a junction capacitance of 13 pF. Assume the junction potential is $\phi = 0.7 \text{ V}$ and the grading coefficient is $\gamma = 0.33$.

Detailed Solution

5.1 Nyquist oscillation criterion and Colpitts feedback network

Let the loop gain be

$$L(s) = A(s)\beta(s). \quad (165)$$

The closed-loop characteristic equation is

$$1 + L(s) = 0. \quad (166)$$

According to the Nyquist stability criterion, sustained oscillation occurs at the boundary of stability when the Nyquist locus passes through the critical point $-1 + j0$. Equivalently, at the oscillation frequency ω_0 ,

$$|A(j\omega_0)\beta(j\omega_0)| = 1, \quad (167)$$

and

$$\angle A(j\omega_0)\beta(j\omega_0) = (2m + 1)180^\circ \quad (168)$$

for the negative-feedback form $1 + L = 0$. In the usual positive-feedback Barkhausen representation, the total loop phase is 0° or an integer multiple of 360° .

A Colpitts oscillator uses an inductor and two series capacitors as the frequency-selective feedback network. The equivalent capacitance is

$$C_{\text{eq}} = \frac{C_1 C_2}{C_1 + C_2}. \quad (169)$$

The oscillation frequency is approximately

$$f_0 = \frac{1}{2\pi\sqrt{LC_{\text{eq}}}}. \quad (170)$$

The ratio of C_1 and C_2 determines the feedback fraction. Proper selection of the ratio allows the circuit to start reliably without using excess transistor gain or bias current.

From an energy-efficiency and environmental perspective, the following design choices are beneficial:

- (a) High- Q inductors and low-loss capacitors reduce circulating loss and lower the required active-device power.
- (b) Correct feedback ratio avoids overdriving the transistor, reducing harmonic generation and wasted DC power.
- (c) A stable resonant frequency reduces retransmissions and spectral interference in wireless systems.
- (d) Varactor or digitally controlled tuning can replace mechanically adjusted components and reduce maintenance and component replacement.
- (e) Lower power consumption reduces heat generation, cooling demand, and lifetime carbon emissions.

5.2 Oscillation frequency

Given

$$L = 70 \text{ nH}, \quad C_1 = 60 \text{ pF}, \quad C_2 = 350 \text{ pF}, \quad (171)$$

the equivalent capacitance is

$$C_{\text{eq}} = \frac{(60)(350)}{60 + 350} \quad (172)$$

$$= 51.22 \text{ pF}. \quad (173)$$

Therefore,

$$f_0 = \frac{1}{2\pi\sqrt{(70 \times 10^{-9})(51.22 \times 10^{-12})}} \quad (174)$$

$$= 8.405 \times 10^7 \text{ Hz} \quad (175)$$

$$= 84.05 \text{ MHz}. \quad (176)$$

Hence,

$$\boxed{f_0 \approx 84.05 \text{ MHz}.} \quad (177)$$

5.3 Capacitor C_1 from transistor gain

Using the common Colpitts start-up approximation

$$\beta_{\text{transistor}} \geq \frac{C_2}{C_1}, \quad (178)$$

the limiting design value is

$$C_1 = \frac{C_2}{\beta_{\text{transistor}}}. \quad (179)$$

Given

$$C_2 = 7 \text{ nF}, \quad \beta_{\text{transistor}} = 45, \quad (180)$$

$$C_1 = \frac{7 \text{ nF}}{45} \quad (181)$$

$$= 0.1556 \text{ nF} \quad (182)$$

$$= 155.6 \text{ pF}. \quad (183)$$

Therefore,

$$\boxed{C_1 \approx 156 \text{ pF}.} \quad (184)$$

The exact capacitor ratio depends on the transistor connection and on which capacitor is defined as the feedback capacitor. The expression above uses the standard convention stated in the solution.

5.4 Required inductance at 30 MHz

Given

$$f_0 = 30 \text{ MHz}, \quad C_1 = 170 \text{ pF}, \quad C_2 = 190 \text{ pF}, \quad (185)$$

the equivalent capacitance is

$$C_{\text{eq}} = \frac{(170)(190)}{170 + 190} \quad (186)$$

$$= 89.72 \text{ pF}. \quad (187)$$

From

$$f_0 = \frac{1}{2\pi\sqrt{LC_{\text{eq}}}}, \quad (188)$$

the inductance is

$$L = \frac{1}{(2\pi f_0)^2 C_{\text{eq}}}. \quad (189)$$

Therefore,

$$L = \frac{1}{[2\pi(30 \times 10^6)]^2(89.72 \times 10^{-12})} \quad (190)$$

$$= 3.137 \times 10^{-7} \text{ H} \quad (191)$$

$$= 313.7 \text{ nH}. \quad (192)$$

Hence,

$$\boxed{L \approx 314 \text{ nH}.} \quad (193)$$

5.5 Reverse bias required for a graded-junction diode

The junction capacitance is

$$C_j(V_R) = \frac{C_0}{\left(1 + \frac{V_R}{\phi}\right)^\gamma}. \quad (194)$$

Rearranging,

$$V_R = \phi \left[\left(\frac{C_0}{C_j} \right)^{1/\gamma} - 1 \right]. \quad (195)$$

Given

$$C_0 = 22 \text{ pF}, \quad C_j = 13 \text{ pF}, \quad \phi = 0.7 \text{ V}, \quad \gamma = 0.33, \quad (196)$$

$$V_R = 0.7 \left[\left(\frac{22}{13} \right)^{1/0.33} - 1 \right] \quad (197)$$

$$= 2.75 \text{ V}. \quad (198)$$

Therefore,

$$\boxed{V_R \approx 2.75 \text{ V}.} \quad (199)$$

6 Question 6: Varactor-Tuned Resonant Circuits

Original Question

Part A

A varactor-tuned LC resonant circuit operates at 6 MHz with $L = 65 \mu\text{H}$. It must be tunable over a $\pm 3\%$ frequency range using a single silicon graded-junction varactor diode with $\gamma = 0.5$. Assume ideal components.

- (i) Calculate the required capacitance range.
- (ii) Determine the corresponding varactor capacitance at both tuning limits.

Part B

A varactor diode with

$$C_0 = 30 \text{ pF}, \quad \phi = 0.7 \text{ V}, \quad \gamma = 0.33$$

is added in parallel with a resonant circuit operating at 5 MHz, with $L = 53 \mu\text{H}$.

- (i) Create a design that reduces the resonant frequency by 3% using reverse bias.
- (ii) Calculate the required DC bias voltage applied to the varactor.

Part C

An existing RF tuning circuit uses manual trimmer capacitors. Redesign the tuning approach to better satisfy zero-carbon and environmental objectives, and evaluate the benefits and trade-offs.

Detailed Solution

6.1 Capacitance range for a $\pm 3\%$ tuning range

The nominal frequency is

$$f_0 = 6 \text{ MHz}. \quad (200)$$

The lower and upper frequencies are

$$f_{\min} = 0.97f_0 = 5.82 \text{ MHz}, \quad (201)$$

$$f_{\max} = 1.03f_0 = 6.18 \text{ MHz}. \quad (202)$$

For an ideal LC resonator,

$$C = \frac{1}{(2\pi f)^2 L}. \quad (203)$$

Given $L = 65 \mu\text{H}$, the maximum capacitance occurs at the minimum frequency:

$$C_{\max} = \frac{1}{[2\pi(5.82 \times 10^6)]^2(65 \times 10^{-6})} \quad (204)$$

$$= 11.505 \text{ pF}. \quad (205)$$

The minimum capacitance occurs at the maximum frequency:

$$C_{\min} = \frac{1}{[2\pi(6.18 \times 10^6)]^2(65 \times 10^{-6})} \quad (206)$$

$$= 10.204 \text{ pF}. \quad (207)$$

Therefore,

$$\boxed{C_{\min} = 10.204 \text{ pF}, \quad C_{\max} = 11.505 \text{ pF}.} \quad (208)$$

The required tuning span is

$$\Delta C = C_{\max} - C_{\min} \quad (209)$$

$$= 1.301 \text{ pF}. \quad (210)$$

Hence,

$$\boxed{\Delta C \approx 1.30 \text{ pF}.} \quad (211)$$

For a single ideal varactor that provides all of the resonant capacitance,

$$\boxed{C_V(f_{\min}) = 11.505 \text{ pF}, \quad C_V(f_{\max}) = 10.204 \text{ pF}.} \quad (212)$$

6.2 Reduction of a 5 MHz resonant frequency by 3%

The original frequency is

$$f_0 = 5 \text{ MHz}, \quad (213)$$

and the target frequency is

$$f_1 = 0.97f_0 \quad (214)$$

$$= 4.85 \text{ MHz}. \quad (215)$$

With $L = 53 \text{ }\mu\text{H}$, the original capacitance is

$$C_{\text{original}} = \frac{1}{[2\pi(5 \times 10^6)]^2(53 \times 10^{-6})} \quad (216)$$

$$= 19.117 \text{ pF}. \quad (217)$$

The required total capacitance at 4.85 MHz is

$$C_{\text{new}} = \frac{1}{[2\pi(4.85 \times 10^6)]^2(53 \times 10^{-6})} \quad (218)$$

$$= 20.318 \text{ pF}. \quad (219)$$

Therefore, the varactor must add

$$C_V = C_{\text{new}} - C_{\text{original}} \quad (220)$$

$$= 20.318 - 19.117 \quad (221)$$

$$= 1.201 \text{ pF}. \quad (222)$$

Hence,

$$\boxed{C_V \approx 1.201 \text{ pF}.} \quad (223)$$

For the varactor,

$$C_V = \frac{C_0}{\left(1 + \frac{V_R}{\phi}\right)^\gamma}. \quad (224)$$

Solving for V_R ,

$$V_R = \phi \left[\left(\frac{C_0}{C_V} \right)^{1/\gamma} - 1 \right]. \quad (225)$$

Using

$$C_0 = 30 \text{ pF}, \quad \phi = 0.7 \text{ V}, \quad \gamma = 0.33, \quad (226)$$

$$V_R = 0.7 \left[\left(\frac{30}{1.201} \right)^{1/0.33} - 1 \right] \quad (227)$$

$$\approx 1.20 \cdot 10^4 \text{ V}. \quad (228)$$

Thus, the direct mathematical result is

$$\boxed{V_R \approx 12.0 \text{ kV}.} \quad (229)$$

This result is physically impractical. It shows that placing the stated 30 pF varactor directly in parallel is not a suitable implementation because the required effective capacitance is only about 1.2 pF. A practical circuit should place a small fixed capacitor in series with the varactor so that the series equivalent capacitance is approximately 1.2 pF, while the varactor itself operates at a safe reverse bias.

6.3 Environmentally improved tuning approach

The manual trimmer capacitor may be replaced by an electronically controlled varactor-tuning network. The proposed design contains:

- (i) a reverse-biased varactor diode,
- (ii) a low-power DAC or PWM-to-DC control circuit,
- (iii) a high-value RF isolation resistor or RF choke,

- (iv) a DC-blocking capacitor where required,
- (v) closed-loop frequency calibration in firmware.

The principal benefits are:

- (a) No repeated mechanical adjustment is required.
- (b) Remote calibration reduces travel and maintenance.
- (c) Automatic tuning can compensate for temperature and component ageing.
- (d) Improved matching can reduce transmitter power loss and retransmission.
- (e) Reduced mechanical wear may extend product lifetime and reduce electronic waste.

The trade-offs are:

- (a) A varactor has finite Q , causing additional RF loss.
- (b) The capacitance-voltage relationship is nonlinear.
- (c) Bias noise can cause phase noise or frequency modulation.
- (d) The tuning range may be smaller than that of a mechanical trimmer.
- (e) Additional control electronics create embodied carbon and require a small amount of operating power.

Overall, electronic tuning is more consistent with zero-carbon objectives when the circuit is produced at scale, operates for a long lifetime, and reduces maintenance, power loss, and premature replacement.

6.4 Proposed varactor-tuning circuit

A practical electronically controlled tuning network is shown in Figure 4. The fixed capacitor C_F establishes the main resonant capacitance, while the series capacitor C_S limits the effective contribution of the varactor. The large bias resistor R_B isolates the RF tank from the DC tuning source, and C_B bypasses noise on the tuning voltage.

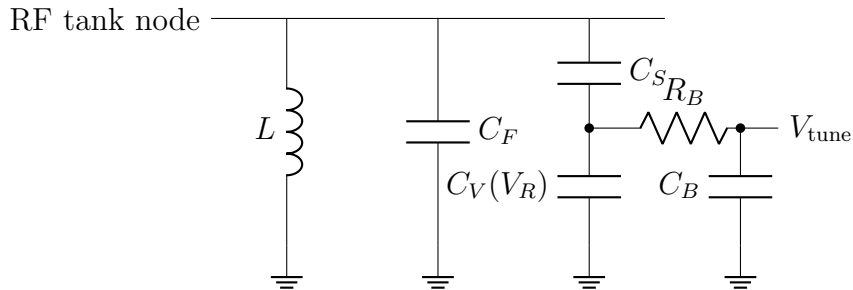


Figure 4: Electronically controlled varactor-tuning network.

At the RF frequency, R_B is selected to be much larger than the tank impedance so that the control circuit does not significantly load the resonator. The bypass capacitor C_B should present a low impedance to ground at unwanted noise and modulation frequencies.

7 Question 7: Simultaneous Conjugate Matching

Original Question

A 300 MHz silicon transistor has the following S -parameters at $V_{CE} = 10\text{ V}$ and $I_C = 10\text{ mA}$:

$$[S] = \begin{bmatrix} 0.39\angle 160^\circ & 0.02\angle 58^\circ \\ 4.9\angle 62^\circ & 0.37\angle (-42^\circ) \end{bmatrix}. \quad (230)$$

The amplifier operates between $50\ \Omega$ terminations. Design the input and output matching networks for simultaneous conjugate matching and maximum gain by calculating:

- (i) Rollet's stability factor K and the determinant Δ ;
- (ii) the maximum available gain;
- (iii) the load reflection coefficient Γ_L ;
- (iv) the source reflection coefficient Γ_S ;
- (v) the corresponding input-side impedance;
- (vi) the corresponding output-side impedance.

Detailed Solution

The transistor scattering parameters are

$$[S] = \begin{bmatrix} 0.39\angle 160^\circ & 0.02\angle 58^\circ \\ 4.9\angle 62^\circ & 0.37\angle -42^\circ \end{bmatrix}. \quad (231)$$

The system reference impedance is

$$Z_0 = 50\ \Omega. \quad (232)$$

7.1 Rollet stability factor and determinant

The determinant is

$$\Delta = S_{11}S_{22} - S_{12}S_{21}. \quad (233)$$

Converting and calculating,

$$\Delta = -0.01874 + j0.04254 \quad (234)$$

$$= 0.04649\angle 113.78^\circ. \quad (235)$$

Therefore,

$$\boxed{\Delta = -0.01874 + j0.04254, \quad |\Delta| = 0.04649.} \quad (236)$$

Rollet's stability factor is

$$K = \frac{1 - |S_{11}|^2 - |S_{22}|^2 + |\Delta|^2}{2|S_{12}S_{21}|}. \quad (237)$$

Substitution gives

$$K = \frac{1 - (0.39)^2 - (0.37)^2 + (0.04649)^2}{2(0.02)(4.9)} \quad (238)$$

$$= 3.639. \quad (239)$$

Therefore,

$$\boxed{K = 3.639.} \quad (240)$$

Since

$$K > 1 \quad \text{and} \quad |\Delta| < 1, \quad (241)$$

the transistor is unconditionally stable at the stated frequency and bias.

7.2 Maximum available gain

For an unconditionally stable transistor,

$$MAG = \left| \frac{S_{21}}{S_{12}} \right| \left(K - \sqrt{K^2 - 1} \right). \quad (242)$$

Thus,

$$MAG = \frac{4.9}{0.02} \left(3.639 - \sqrt{3.639^2 - 1} \right) \quad (243)$$

$$= 34.33. \quad (244)$$

In decibels,

$$MAG_{\text{dB}} = 10 \log_{10}(34.33) \quad (245)$$

$$= 15.36 \text{ dB}. \quad (246)$$

Hence,

$$\boxed{MAG = 34.33 = 15.36 \text{ dB}.} \quad (247)$$

7.3 Optimum source and load reflection coefficients

Define

$$B_1 = 1 + |S_{11}|^2 - |S_{22}|^2 - |\Delta|^2, \quad (248)$$

$$C_1 = S_{11} - \Delta S_{22}^*, \quad (249)$$

$$B_2 = 1 + |S_{22}|^2 - |S_{11}|^2 - |\Delta|^2, \quad (250)$$

and

$$C_2 = S_{22} - \Delta S_{11}^*. \quad (251)$$

Numerically,

$$B_1 = 1.01304, \quad (252)$$

$$C_1 = -0.35079 + j0.12633, \quad (253)$$

$$B_2 = 0.98264, \quad (254)$$

and

$$C_2 = 0.26242 - j0.23449. \quad (255)$$

The optimum source reflection coefficient is

$$\Gamma_S = \frac{B_1 - \sqrt{B_1^2 - 4|C_1|^2}}{2C_1}, \quad (256)$$

where the sign is selected so that $|\Gamma_S| < 1$.

Thus,

$$\Gamma_S = -0.41301 - j0.14874 \quad (257)$$

$$= 0.43897 \angle -160.19^\circ. \quad (258)$$

Therefore,

$$\boxed{\Gamma_S = -0.4130 - j0.1487 = 0.4390 \angle -160.19^\circ.} \quad (259)$$

Similarly,

$$\Gamma_L = \frac{B_2 - \sqrt{B_2^2 - 4|C_2|^2}}{2C_2}. \quad (260)$$

Hence,

$$\Gamma_L = 0.31459 + j0.28110 \quad (261)$$

$$= 0.42188 \angle 41.78^\circ. \quad (262)$$

Therefore,

$$\boxed{\Gamma_L = 0.3146 + j0.2811 = 0.4219 \angle 41.78^\circ.} \quad (263)$$

7.4 Corresponding source and load impedances

The impedance corresponding to a reflection coefficient is

$$Z = Z_0 \frac{1 + \Gamma}{1 - \Gamma}. \quad (264)$$

For the source,

$$Z_S = 50 \frac{1 + \Gamma_S}{1 - \Gamma_S} \quad (265)$$

$$= 19.996 - j7.368 \, \Omega. \quad (266)$$

Thus,

$$\boxed{Z_S \approx 20.00 - j7.37 \, \Omega.} \quad (267)$$

For the load,

$$Z_L = 50 \frac{1 + \Gamma_L}{1 - \Gamma_L} \quad (268)$$

$$= 74.891 + j51.221 \, \Omega. \quad (269)$$

Thus,

$$\boxed{Z_L \approx 74.89 + j51.22 \, \Omega.} \quad (270)$$

The input matching network must transform the external $50 \, \Omega$ source to Z_S , while the output matching network must transform the external $50 \, \Omega$ load to Z_L .

7.5 Input and output matching-network layout

The matching-network arrangement is shown in Figure 5. The input network presents the optimum source impedance to port 1 of the transistor, and the output network presents the optimum load impedance to port 2.

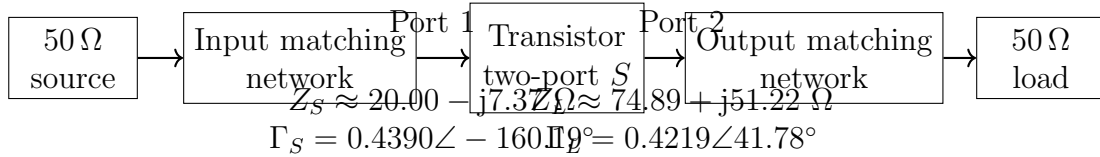


Figure 5: Simultaneous-conjugate matching arrangement for Question 7.

The matching blocks may be implemented using lumped L -networks, single-stub networks, or microstrip sections. Their purpose is the same: to convert the external $50 \, \Omega$ terminations into the calculated complex impedances at the transistor reference planes.

8 Question 8: Power-Amplifier Stability and Matching

Original Question

A power amplifier must deliver 10 W (40) output power at 3 GHz, with system impedance $Z_0 = 50 \Omega$. The following small-signal S -parameters are given at 3 GHz and $V_{DS} = 25$ V:

$$S_{11} = 0.32\angle(-150^\circ), \quad S_{12} = 0.004\angle 50^\circ,$$

$$S_{21} = 4.5\angle(-90^\circ), \quad S_{22} = 0.36\angle(-120^\circ).$$

Large-signal load-pull data for 10 W operation give

$$Z_{SP} = 17 - j5 \Omega, \quad Z_{LP} = 6 + j5 \Omega.$$

- Determine the stability of the amplifier at 2.4 GHz using the supplied small-signal scattering parameters.
- Evaluate the effect of unconditional stability on public health and safety.
- Convert the large-signal source and load impedances into reflection coefficients.
- Determine the source and load reflection coefficients for simultaneous conjugate matching and calculate their corresponding impedances.
- If the drain efficiency is $\eta_D = 28\%$, determine:
 - the maximum available gain;
 - the power-added efficiency.

Detailed Solution

The small-signal parameters supplied are

$$S_{11} = 0.32\angle -150^\circ, \tag{271}$$

$$S_{12} = 0.004\angle 50^\circ, \tag{272}$$

$$S_{21} = 4.5\angle -90^\circ, \tag{273}$$

and

$$S_{22} = 0.36\angle -120^\circ. \tag{274}$$

The data are stated at 3 GHz, although part (a) asks for stability at 2.4 GHz. Since no separate 2.4 GHz data are supplied, the following calculation uses the given S -parameters and therefore strictly represents the stated 3 GHz operating point.

8.1 Stability

The determinant is

$$\Delta = S_{11}S_{22} - S_{12}S_{21} \quad (275)$$

$$= -0.01379 + j0.12677 \quad (276)$$

$$= 0.12752 \angle 96.21^\circ. \quad (277)$$

Therefore,

$$\boxed{|\Delta| = 0.1275.} \quad (278)$$

The stability factor is

$$K = \frac{1 - |S_{11}|^2 - |S_{22}|^2 + |\Delta|^2}{2|S_{12}S_{21}|} \quad (279)$$

$$= 21.785. \quad (280)$$

Thus,

$$\boxed{K = 21.785.} \quad (281)$$

Since

$$K > 1 \quad \text{and} \quad |\Delta| < 1, \quad (282)$$

the device is unconditionally stable for all passive source and load terminations at the frequency represented by the supplied data.

8.2 Effect of unconditional stability on health and safety

Unconditional stability reduces the risk of unintended self-oscillation when the amplifier is connected to passive source and load impedances. This has several safety implications:

- (a) It reduces unintended RF emissions that may interfere with medical, aviation, navigation, and communication systems.
- (b) It reduces excessive transistor current and overheating caused by parasitic oscillation.
- (c) It lowers the risk of component failure, burns, smoke, or fire in high-power RF hardware.
- (d) It improves compliance with electromagnetic compatibility and occupational RF-exposure limits.
- (e) It improves reliability under antenna mismatch, cable movement, and manufacturing tolerances.

Unconditional small-signal stability does not by itself guarantee complete large-signal safety. Bias-network resonances, thermal instability, out-of-band oscillations, and breakdown under severe mismatch must still be checked.

8.3 Large-signal load-pull reflection coefficients

The load-pull source impedance is

$$Z_{SP} = 17 - j5 \, \Omega. \quad (283)$$

The corresponding reflection coefficient is

$$\Gamma_{SP} = \frac{Z_{SP} - Z_0}{Z_{SP} + Z_0} \quad (284)$$

$$= \frac{(17 - j5) - 50}{(17 - j5) + 50} \quad (285)$$

$$= -0.48427 - j0.11077 \quad (286)$$

$$= 0.49678 \angle -167.12^\circ. \quad (287)$$

Therefore,

$$\boxed{\Gamma_{SP} = -0.4843 - j0.1108 = 0.4968 \angle -167.12^\circ.} \quad (288)$$

The load-pull load impedance is

$$Z_{LP} = 6 + j5 \, \Omega. \quad (289)$$

Thus,

$$\Gamma_{LP} = \frac{Z_{LP} - Z_0}{Z_{LP} + Z_0} \quad (290)$$

$$= \frac{(6 + j5) - 50}{(6 + j5) + 50} \quad (291)$$

$$= -0.77159 + j0.15818 \quad (292)$$

$$= 0.78764 \angle 168.41^\circ. \quad (293)$$

Therefore,

$$\boxed{\Gamma_{LP} = -0.7716 + j0.1582 = 0.7876 \angle 168.41^\circ.} \quad (294)$$

8.4 Small-signal simultaneous conjugate match

Using the simultaneous conjugate-match equations,

$$\Gamma_S = \frac{B_1 - \sqrt{B_1^2 - 4|C_1|^2}}{2C_1}, \quad (295)$$

and

$$\Gamma_L = \frac{B_2 - \sqrt{B_2^2 - 4|C_2|^2}}{2C_2}, \quad (296)$$

the passive roots are

$$\Gamma_S = -0.27597 + j0.15274 \quad (297)$$

$$= 0.31542 \angle 151.04^\circ, \quad (298)$$

and

$$\Gamma_L = -0.18228 + j0.30588 \quad (299)$$

$$= 0.35607 \angle 120.79^\circ. \quad (300)$$

Therefore,

$$\boxed{\Gamma_S = -0.2760 + j0.1527 = 0.3154 \angle 151.04^\circ}, \quad (301)$$

and

$$\boxed{\Gamma_L = -0.1823 + j0.3059 = 0.3561 \angle 120.79^\circ}. \quad (302)$$

The corresponding impedances are

$$Z_S = 50 \frac{1 + \Gamma_S}{1 - \Gamma_S} \quad (303)$$

$$= 27.265 + j9.249 \, \Omega, \quad (304)$$

and

$$Z_L = 50 \frac{1 + \Gamma_L}{1 - \Gamma_L} \quad (305)$$

$$= 29.276 + j20.510 \, \Omega. \quad (306)$$

Hence,

$$\boxed{Z_S \approx 27.26 + j9.25 \, \Omega}, \quad (307)$$

and

$$\boxed{Z_L \approx 29.28 + j20.51 \, \Omega}. \quad (308)$$

The small-signal conjugate-match impedances are different from the large-signal load-pull impedances because a power amplifier is nonlinear at the 10 W operating point. For maximum output power and efficiency, the load-pull impedances normally take priority.

8.5 Maximum available gain

Because the amplifier is unconditionally stable,

$$MAG = \left| \frac{S_{21}}{S_{12}} \right| \left(K - \sqrt{K^2 - 1} \right). \quad (309)$$

Thus,

$$MAG = \frac{4.5}{0.004} \left(21.785 - \sqrt{21.785^2 - 1} \right) \quad (310)$$

$$= 25.834. \quad (311)$$

In decibels,

$$MAG_{\text{dB}} = 10 \log_{10}(25.834) \quad (312)$$

$$= 14.12 \text{ dB}. \quad (313)$$

Therefore,

$$\boxed{MAG = 25.83 = 14.12 \text{ dB}.} \quad (314)$$

8.6 Power-added efficiency

The output power is

$$P_{\text{out}} = 10 \text{ W}. \quad (315)$$

The drain efficiency is

$$\eta_D = \frac{P_{\text{out}}}{P_{\text{DC}}} = 0.28. \quad (316)$$

Therefore,

$$P_{\text{DC}} = \frac{P_{\text{out}}}{\eta_D} \quad (317)$$

$$= \frac{10}{0.28} \quad (318)$$

$$= 35.714 \text{ W}. \quad (319)$$

Assuming the available gain is used as the power gain,

$$P_{\text{in}} = \frac{P_{\text{out}}}{MAG} \quad (320)$$

$$= \frac{10}{25.834} \quad (321)$$

$$= 0.387 \text{ W}. \quad (322)$$

The power-added efficiency is

$$PAE = \frac{P_{\text{out}} - P_{\text{in}}}{P_{\text{DC}}}. \quad (323)$$

Thus,

$$PAE = \frac{10 - 0.387}{35.714} \tag{324}$$

$$= 0.2692 \tag{325}$$

$$= 26.92\%. \tag{326}$$

Therefore,

$$\boxed{PAE \approx 26.92\%}. \tag{327}$$